

SHIVAIAH VARI AMRUTH RAO

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PROFILE SUMMARY

Final-year B.Tech Computer Science and Engineering student (Class of 2027, CGPA 7.6, no active backlogs) with strong coding and problem-solving skills across Java, Python, JavaScript, and C/C++. Hands-on experience designing and integrating full-stack applications and REST APIs using React.js, Node.js, and MongoDB, with foundational cloud computing knowledge (NPTEL-certified). Self-motivated, quick to learn new platforms, and a strong communicator, seeking to implement and customize Oracle Cloud Applications (CRM, ERP, HCM) and build integrations using enterprise PaaS tools.

TECHNICAL SKILLS

- Programming Languages: Java, Python, JavaScript (ES6+), C, C++
- Core CS Domain Knowledge: Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Cloud & DevOps fundamentals
- Cloud & Systems Integration: NPTEL Cloud Computing (Certified); REST API design and integration; enterprise system integration experience through full-stack project delivery
- Databases & Backend: MongoDB, MySQL, Node.js, Express
- Applied ML/AI: Python, TensorFlow, OpenCV, NLP fundamentals, model evaluation and output analysis
- Tools & Platforms: Git/GitHub, VS Code; AI-assisted development tools (Claude, GitHub Copilot, Cursor, Windsurf)

PROJECTS

RECALL — Modern Learning Hub for Students and Institutions *(React.js, Node.js, MongoDB)*

[Live: recall-app-umber.vercel.app](https://recall-app-umber.vercel.app)

- Built and deployed a centralized web platform modeling prerequisite relationships across 44 topics via a custom knowledge-dependency graph, reducing redundant topic review time by 25%.

Sportsync — Sports Facility Booking Platform *(React.js, Node.js, Express, MongoDB)*

[Live: sportsync-d644b.web.app](https://sportsync-d644b.web.app)

- Designed and rigorously tested a full frontend-backend REST API integration, reducing real-time slot-booking conflicts by 35% by systematically identifying and resolving logic errors in availability handling.
- Reviewed and validated authentication flows against expected behavior, cutting authentication-related bugs by 40% through structured, iterative testing across 15+ test cases.

Nexora — Developer Skill Assessment & Competitive Coding Platform *(Next.js 16)*

[Live: nexora-iygu.vercel.app](https://nexora-iygu.vercel.app)

- Built a full-stack developer skill verification platform with an interactive Monaco IDE, automated test evaluation, real-time leaderboards, and verifiable developer profiles; achieved <200ms code execution latency across 4 programming languages using Judge0/Docker sandboxing and secured 100% of database access with PostgreSQL Row-Level Security (RLS) through Supabase.

Virtual Drug Discovery with QML *(React.js, Vite)*

[Live: virtual-drug-discovery.web.app](https://virtual-drug-discovery.web.app)

- Built an interactive frontend to visualize model confidence scores across 50+ compound predictions, restructuring data flow between components to improve load performance by 30% across devices.

Edupath-AI — Personalised Learning Companion *(React.js — In Progress)*

- Building an adaptive learning platform using reusable, state-driven components, cutting UI code duplication by 20% through modular design; testing and refining logic based on iterative feedback.

Drowsiness Detection System *(Python, OpenCV)*

- Developed a real-time driver-alert system using Haar cascade classifiers, achieving 92% detection accuracy at 15+ FPS through iterative threshold tuning on live video input.

EDUCATION

B.Tech — Computer Science and Engineering, MLR Institute of Technology

2023 – 2027

CGPA: 7.6 / 10

Intermediate (MPC), Narayana Junior College

2021 – 2023

78.2%

SSC — Dr. K.K.R. Gowtham Concept School

2020 – 2021

GPA: 10.0 / 10

CERTIFICATIONS

- NPTEL — Cloud Computing
- HackerRank — Software Engineer (Certified)
- HackerRank — Software Engineer Intern (Certified)

KEY STRENGTHS

- Strong Coder & Problem-Solver — applies algorithmic and CS reasoning to write, debug, and evaluate solutions across multiple languages.
- Self-Motivated & Quick Learner — comfortable picking up new platforms and enterprise tools independently, working against structured requirements.
- Systems Integrator — experience connecting frontend, backend, and database layers via REST APIs to deliver seamless business processes.
- Effective Communicator — documents technical reasoning clearly for both technical and non-technical audiences.